

Unifying Special Relativity and Quantum Mechanics

Lecture 15 — The Minimal Coupling Prescription

(Gauging a Global Symmetry, the Covariant Derivative, and Scalar QED)

Typed version of the handwritten notes, with added explanations in blue boxes

How to read this document

Conventions. Exactly as in Lectures 1–14: everything in the normal text and in the **yellow** / **pink** / **grey** boxes is a faithful transcription of the handwritten notes (the coloured boxes reproduce the results that were framed or highlighted in the original — yellow boxes for the yellow-highlighted results, pink boxes for the pink/red-framed ones, grey boxes for the plain pencil frames). The colours of the text roughly follow the colours of the pens used in the original. Everything inside a **blue box like this one** is extra material that I added: derivations spelled out, terminology, corrections of transcription slips, figures, and connections to earlier lectures.

Pagination of this scan. This lecture opens a *new* chapter: the page numbering restarts at 1 (this is the **eighth restart** in the collection of scans). The sheets are therefore marked [p. 1*****]–[p. 18*****], with **eight stars**.

Good news: this scan has no anomalies. Unlike Lectures 3, 6–9 and 13–14, the eighteen sheets of this lecture are complete, in order, and none is duplicated. The page markers can be trusted literally.

Where we left off. Lecture 14 built the massive spin-1 (Proca) field, showed that the massless limit $M \rightarrow 0$ exists *if and only if* the vector couples to a conserved current $\partial_\mu j^\mu = 0$, and ended with the Lagrangian $\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}M^2V_\mu V^\mu - V_\mu j^\mu$, under the standing assumption that j^μ does not depend on V^μ . This lecture takes that assumption apart. It asks the question that was left hanging: *where does a conserved current actually come from, and what guarantees it stays conserved once the interaction is switched on?* The answer — gauging a global $U(1)$ symmetry, and the covariant derivative that it forces on us — is the single most important construction of the whole course.

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Part I — Two limitations, and two examples

(pp. 1*****-5*****)

1 The minimal coupling prescription: the two limitations [p. 1*****]

The previous discussion has some limitations.

1) In the Lagrangian

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}M^2V_\mu V^\mu - V_\mu J^\mu$$

we assumed that J^μ does not depend on the field V_μ or its derivatives.

2) Suppose that we really want to implement the strategy of “coupling the photon with a conserved current”.

It is not clear how to construct such object J^μ .

Let us consider the following examples.

Added explanation (not in the original notes)

A change of letter, not of physics. The Lagrangian above is literally the last Lagrangian of Lecture 14: the massive vector field is still called V^μ , while its source — written j^μ in Lecture 14 — is now written J^μ . From the next sheet on, the notes drop V^μ and write A^μ throughout, because we are heading straight for the *photon*. So: V^μ and A^μ denote the same field, and j^μ and J^μ the same current; only the letters change.

What the two limitations really say. They are two halves of the same complaint.

- Limitation 1 is a *technical* restriction: all of Lecture 14’s manipulations (the equation of motion, the integration by parts that saved the longitudinal mode) assumed $\partial\mathcal{L}/\partial A_\mu = -J^\mu$ with J^μ blind to A_μ . We will discover on p. 17***** that in *scalar* QED this is simply false: the conserved current contains A^μ .
- Limitation 2 is a *conceptual* one: “take a conserved current and couple it to the photon” is a recipe with no supplier. A current conserved in the *free* theory need not stay conserved once you have added the coupling that changes the equations of motion — and it is the *interacting* conservation law that gauge invariance requires.

The whole lecture is the resolution: stop guessing currents, and instead *gauge* the global symmetry. The current, the coupling, and the conservation law then all come out together, forced.

2 Example (i): the free Dirac theory [p. 1*****-2*****]

① Take the free theory

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\Psi}(i\gamma^\mu\partial_\mu - m)\Psi$$

There is a **global** $U(1)$ **symmetry** under which $\Psi \rightarrow \exp(-iq\theta)\Psi$, with current

$$J^\mu = q\bar{\Psi}\gamma^\mu\Psi$$

conserved on the E.o.M.

The check is simple. The E.o.M. are

$$\begin{cases} i\gamma^\mu\partial_\mu\Psi - m\Psi = 0 \\ (-i)(\partial_\mu\bar{\Psi})\gamma^\mu - m\bar{\Psi} = 0 \end{cases} \implies \begin{cases} \gamma^\mu(\partial_\mu\Psi) = -im\Psi \\ (\partial_\mu\bar{\Psi})\gamma^\mu = +im\bar{\Psi} \end{cases}$$

$$\begin{aligned}\partial_\mu J^\mu &= q \partial_\mu (\bar{\Psi} \gamma^\mu \Psi) = q \left[(\partial_\mu \bar{\Psi}) \gamma^\mu \Psi + \bar{\Psi} \gamma^\mu (\partial_\mu \Psi) \right] \quad (\text{using the E.o.M.}) \\ &= q \left[im \bar{\Psi} \Psi + \bar{\Psi} (-im \Psi) \right] = 0\end{aligned}$$

It seems, therefore, natural to construct the coupling with the J^μ defined before

$$\rightsquigarrow \mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \bar{\Psi} (i\gamma^\mu \partial_\mu - m) \Psi - q A_\mu \bar{\Psi} \gamma^\mu \Psi$$

and indeed, it works.

There is, however, a subtle point. Since we changed the theory, the E.o.M. of Ψ changed, and we should check that $\partial_\mu J^\mu = 0$ **on the E.o.M. of the interacting theory**.

Let's compute.

$$\begin{aligned}\mathcal{L} &= -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \bar{\Psi} (i\gamma^\mu \partial_\mu - m) \Psi - q A_\mu \bar{\Psi} \gamma^\mu \Psi \\ \frac{\partial \mathcal{L}}{\partial (\partial_\mu \bar{\Psi})} &= 0; \quad \frac{\partial \mathcal{L}}{\partial \bar{\Psi}} = i\gamma^\mu \partial_\mu \Psi - m \Psi - q A_\mu \gamma^\mu \Psi\end{aligned}$$

$$\text{E.o.M.} \quad i\gamma^\mu (\partial_\mu \Psi) - m \Psi - q A_\mu \gamma^\mu \Psi = 0$$

E.o.M. of Ψ in the interacting theory.

3 Example (i), continued: the current survives [p. 3*****]

That we also write as

$$\begin{aligned}\gamma^\mu (\partial_\mu \Psi) &= -im \Psi - iq A_\mu \gamma^\mu \Psi \\ (\partial_\mu \bar{\Psi}) \gamma^\mu &= +im \bar{\Psi} + iq \bar{\Psi} \gamma^\mu A_\mu\end{aligned}$$

$$\begin{aligned}\partial_\mu J^\mu &= q \left[(\partial_\mu \bar{\Psi}) \gamma^\mu \Psi + \bar{\Psi} \gamma^\mu (\partial_\mu \Psi) \right] \\ &= q \left[im \bar{\Psi} \Psi + \cancel{iq \bar{\Psi} \gamma^\mu A_\mu \Psi} - im \bar{\Psi} \Psi - \cancel{iq A_\mu \bar{\Psi} \gamma^\mu \Psi} \right] = 0\end{aligned}$$

In this case, the current J^μ is still conserved on the E.o.M. of the fully interacting theory.

↪ **This is, however, a peculiarity of QED and not a generic result.**

Added explanation (not in the original notes)

Why spinor QED gets away with it. Look at what makes the cancellation work. The current $J^\mu = q \bar{\Psi} \gamma^\mu \Psi$ has two special features:

- (a) it contains *no derivatives* of the fields, and
- (b) it contains *no* A_μ .

Because of (a), the interaction term $-q A_\mu \bar{\Psi} \gamma^\mu \Psi$ does not touch $\partial \mathcal{L} / \partial (\partial_\mu \Psi)$, so the Noether current of the *interacting* theory is the *same expression* as the free one. Because of (b), the extra piece that the interaction inserts into the E.o.M. enters $\partial_\mu J^\mu$ twice with opposite signs (once from Ψ , once from $\bar{\Psi}$) and cancels identically.

Neither feature is generic. A scalar field's current *does* contain derivatives — which is exactly why example (ii) is about to fail. Deeper reason: the Dirac Lagrangian is *first order* in derivatives, so its Noether current has no derivatives at all; the Klein–Gordon Lagrangian is second order, so its current does.

One transcription remark. In the middle line of the computation on p. 2***** the overall factor q is dropped and then reinstated; I have kept it throughout for uniformity. Nothing in the argument depends on it, since $q \neq 0$.

4 Example (ii): the charged scalar field [p. 3*****–4*****]

Ⓐ As a second example, we take as free theory

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \underbrace{(\partial_\mu\Phi^\dagger)(\partial^\mu\Phi) - m^2\Phi^\dagger\Phi}$$

Charged free scalar field. The theory also has a **global $U(1)$ symmetry** under which

$$\begin{cases} \Phi \rightarrow e^{-iq\theta} \Phi \\ \Phi^\dagger \rightarrow e^{+iq\theta} \Phi^\dagger \end{cases}$$

Consequently, we have a conserved current that we can construct.

Let's do it, by applying the formula

$$J^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \Phi^a)} D\Phi^a$$

↪ In this formula, the sum over “ a ” means that we should sum over **all independent field variables**.

In the case of the charged scalar field, the application of the canonical quantization procedure requires to treat Φ and $\Phi^\dagger$ as independent field variables.

↪ **Physically**, we describe two d.o.f.: scalar particles with charge “ q ” and scalar anti-particles with charge “ $-q$ ”. In the Hamiltonian formalism, therefore, in order to match the correct d.o.f. we need *two* dynamical field variables.

We thus have

$$J^\mu = (D\Phi^\dagger) \frac{\partial \mathcal{L}}{\partial(\partial_\mu \Phi^\dagger)} + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \Phi)} D\Phi$$

Added explanation (not in the original notes)

Reminder: where the Noether formula comes from (Lecture 13). If the Lagrangian is invariant under an infinitesimal transformation $\Phi^a \rightarrow \Phi^a + \varepsilon D\Phi^a$, then

$$0 = \delta \mathcal{L} = \frac{\partial \mathcal{L}}{\partial \Phi^a} D\Phi^a + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \Phi^a)} \partial_\mu (D\Phi^a) \stackrel{\text{E.o.M.}}{=} \partial_\mu \underbrace{\left[\frac{\partial \mathcal{L}}{\partial(\partial_\mu \Phi^a)} D\Phi^a \right]}_{J^\mu},$$

where the Euler–Lagrange equation was used to replace $\partial \mathcal{L} / \partial \Phi^a = \partial_\mu [\partial \mathcal{L} / \partial(\partial_\mu \Phi^a)]$. The symbol $D\Phi^a$ is the *variation* of the field, i.e. the coefficient of the infinitesimal parameter —

not a derivative.

Warning about the letter “D”. From p. 14***** onwards the same letter D will be used for the *covariant derivative* $D_\mu \Phi = \partial_\mu \Phi + iqA_\mu \Phi$. The two objects are unrelated; you tell them apart by the index: $D\Phi$ (no index) is a variation, $D_\mu \Phi$ (with index) is a derivative. This clash is in the original notes; I keep it, but it is worth flagging once.

5 The scalar current, and where the naive coupling breaks [p. 5*****]

The field variations are

$$\begin{aligned}\Phi &\rightarrow e^{-iq\theta} \Phi = \Phi - iq\theta \Phi &\rightsquigarrow& D\Phi = -iq\Phi \\ \Phi^\dagger &\rightarrow e^{+iq\theta} \Phi^\dagger = \Phi^\dagger + iq\theta \Phi^\dagger &\rightsquigarrow& D\Phi^\dagger = +iq\Phi^\dagger\end{aligned}$$

and we have

$$J^\mu = iq\Phi^\dagger(\partial^\mu \Phi) + (\partial^\mu \Phi^\dagger)(-iq\Phi) = iq[\Phi^\dagger(\partial^\mu \Phi) - (\partial^\mu \Phi^\dagger)\Phi]$$

The current is

$$J^\mu = iq[\Phi^\dagger(\partial^\mu \Phi) - (\partial^\mu \Phi^\dagger)\Phi]$$

and it is conserved on the E.o.M. of the free theory

$$\partial_\mu J^\mu = iq[\Phi^\dagger \square \Phi - (\square \Phi^\dagger)\Phi] = iq(-m^2 \Phi^\dagger \Phi + m^2 \Phi^\dagger \Phi) = 0$$

$$\text{E.o.M.} \quad \begin{cases} \square \Phi = -m^2 \Phi \\ \square \Phi^\dagger = -m^2 \Phi^\dagger \end{cases}$$

It seems, therefore, natural to construct the theory

$$\begin{aligned}\mathcal{L} = & -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + (\partial_\mu \Phi^\dagger)(\partial^\mu \Phi) - m^2 \Phi^\dagger \Phi \\ & - A_\mu iq[\Phi^\dagger(\partial^\mu \Phi) - (\partial^\mu \Phi^\dagger)\Phi]\end{aligned}$$

△ However, we now have different E.o.M., and the current is no longer conserved.

Added explanation (not in the original notes)

Let us see the failure explicitly — the notes assert it, and it is worth two lines of algebra, because the size of the failure is exactly the clue that solves the problem.

With $\mathcal{L}_{\text{int}} = -iqA_\mu[\Phi^\dagger \partial^\mu \Phi - (\partial^\mu \Phi^\dagger)\Phi]$ the Euler–Lagrange equation for $\Phi^\dagger$ becomes

$$\square \Phi = -m^2 \Phi - iq(\partial_\mu A^\mu)\Phi - 2iqA^\mu \partial_\mu \Phi,$$

and its conjugate carries the opposite sign of the q -terms. Substituting into $\partial_\mu J^\mu = iq[\Phi^\dagger \square \Phi - (\square \Phi^\dagger)\Phi]$, the mass terms cancel as before, but the new terms *add* instead of cancelling:

$$\partial_\mu J^\mu = 2q^2 \partial_\mu (A^\mu \Phi^\dagger \Phi) \neq 0$$

Contrast this with example (i): there the extra terms came with opposite signs and killed each other; here they come with the same sign and survive.

The clue. The obstruction is a total divergence of something — namely of $2q^2 A^\mu \Phi^\dagger \Phi$. So if we *redefine* the current as

$$\tilde{J}^\mu = iq[\Phi^\dagger \partial^\mu \Phi - (\partial^\mu \Phi^\dagger) \Phi] - 2q^2 A^\mu \Phi^\dagger \Phi,$$

the divergence cancels. Keep that expression in mind: on p. 17***** it will drop out of the minimal coupling prescription *by itself*, together with the extra “seagull” term $q^2 A_\mu A^\mu \Phi^\dagger \Phi$ that must be added to the Lagrangian. Nothing here has to be guessed — but with the “couple to a conserved current” philosophy, it *would* have to be.

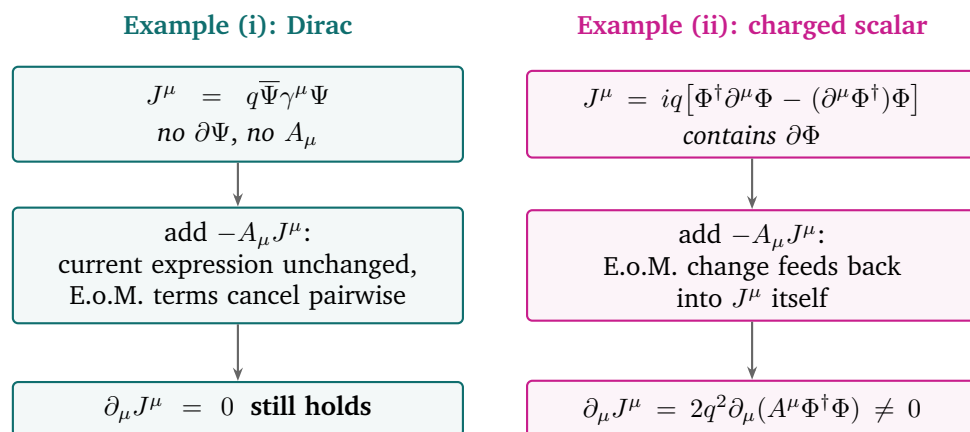


Figure 1. The two examples of pp. 1*****–5***** side by side. Spinor QED works by accident; the scalar case shows that “couple the photon to a conserved current” is not a well-defined prescription. (*Added figure, not in the original notes.*)

Part II — The strategy: gauging the global symmetry

(pp. 6*****–12*****)

6 The strategy: what we know about the full Lagrangian [p. 6*****]

Consider the Lagrangian density

$$\mathcal{L} = \mathcal{L}(\Phi, \partial_\mu \Phi, A^\mu, \partial^\nu A^\mu)$$

This is the Lagrangian of the **full theory** that, at this stage, we do not know.

What do we know about this Lagrangian?

1) Gauge invariance

The physics must remain invariant if we perform a gauge transformation

$$A_\mu(x) \longrightarrow A_\mu(x) + \partial_\mu \alpha(x)$$

 $\hookrightarrow$ The corresponding change in the Lagrangian

$$\begin{aligned} D\mathcal{L} &= \frac{\partial \mathcal{L}}{\partial A^\mu} D A^\mu + \frac{\partial \mathcal{L}}{\partial (\partial^\nu A^\mu)} D \partial^\nu A^\mu \\ &\quad \begin{matrix} \uparrow \\ D A^\mu = \partial^\mu \alpha \end{matrix} \quad \begin{matrix} \uparrow \\ D \partial^\nu A^\mu = \partial^\nu D A^\mu = \partial^\nu \partial^\mu \alpha \end{matrix} \\ &= \frac{\partial \mathcal{L}}{\partial A^\mu} (\partial^\mu \alpha) + \frac{\partial \mathcal{L}}{\partial (\partial^\nu A^\mu)} (\partial^\nu \partial^\mu \alpha) \quad (\star) \end{aligned}$$

must be zero.

2) $U(1)$ global symmetryWe assume that the Lagrangian $\mathcal{L}$ has a $U(1)$ global symmetry under which the field transforms according to [p. 7*****]

$$\Phi(x) \longrightarrow \exp(-iq\theta) \Phi(x)$$

 $\hookrightarrow$ We remark that the presence of this symmetry is **absolutely indispensable** if we want to describe electromagnetic interactions, since particles must be charged.The theory, therefore, admits a $U(1)$ Noether current conserved on the E.o.M. of the full theory.We cannot compute explicitly J^μ , but we can make some considerations.**(2.1)** The Lagrangian is invariant under the transformation

$$\Phi(x) \longrightarrow \Phi(x) - iq\theta \Phi(x)$$

The change in the Lagrangian is

$$\begin{aligned}
 & \mathcal{L}(\Phi - iq\theta\Phi, \partial_\mu\Phi - iq\theta\partial_\mu\Phi) - \mathcal{L}(\Phi, \partial_\mu\Phi) \quad (\text{we only display the } \Phi\text{-dependence}) \\
 &= \cancel{\mathcal{L}(\Phi, \partial_\mu\Phi)} + \frac{\partial\mathcal{L}}{\partial\Phi}(-iq\theta\Phi) + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\Phi)}(-iq\theta)(\partial_\mu\Phi) - \cancel{\mathcal{L}(\Phi, \partial_\mu\Phi)} \\
 &= -iq\theta \left[\frac{\partial\mathcal{L}}{\partial\Phi}\Phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\Phi)}(\partial_\mu\Phi) \right] \stackrel{!}{=} 0
 \end{aligned}$$

We thus have the equation [p. 8*****]

$$\frac{\partial\mathcal{L}}{\partial\Phi}\Phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\Phi)}(\partial_\mu\Phi) = 0 \quad \star$$

(2.2) The general form of the Noether current is

$$J^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\Phi)} D\Phi = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\Phi)} (-iq)\Phi \quad \star\star$$

Added explanation (not in the original notes)

Read $\star$ correctly: it is not an equation of motion. Equation $\star$ is an *algebraic identity* satisfied by the *functional form* of $\mathcal{L}$, valid for arbitrary field configurations, on-shell or off-shell. It just says: “ $\mathcal{L}$ depends on Φ and $\partial_\mu\Phi$ only through $U(1)$ -invariant combinations”. For the free scalar, for instance, $\mathcal{L} \supset (\partial\Phi^\dagger)(\partial\Phi) - m^2\Phi^\dagger\Phi$ and both terms pair a Φ with a $\Phi^\dagger$, so the phases cancel and $\star$ holds identically.

Note also that here (and for the rest of the lecture) only the Φ half of the sum over independent fields is displayed; the $\Phi^\dagger$ half comes along with the conjugate signs, as on p. 4*****. The final scalar-QED formulae on p. 16***** restore both.

Why $\star$ and $\star\star$ are the whole toolkit. These two relations are everything we can extract from a Lagrangian we do not know. $\star$ will be used on the next page to kill the θ -proportional terms; $\star\star$ will be used to *recognise* the leftover as the Noether current. That is the entire trick.

7 **Gauging the symmetry:** $\theta = \theta(x)$ [p. 8*****-9*****]

3) We now do something apparently unjustified.

We assume that the global parameter θ depends on the space-time position “ x ”:

$$\theta = \theta(x)$$

$$\Phi(x) \longrightarrow \exp(-iq\theta(x)) \Phi(x)$$

We “gauge” the global symmetry.

Clearly, we now have $D\mathcal{L} \neq 0$ since $D\mathcal{L} = 0$ only if θ is constant.

So, if $D\mathcal{L} \neq 0$, what is $D\mathcal{L}$ now?

We compute the change in the Lagrangian [p. 9*****]

$$\mathcal{L}(\Phi - iq\theta\Phi, \partial_\mu\Phi - iq\partial_\mu(\theta\Phi)) - \mathcal{L}(\Phi, \partial_\mu\Phi)$$

Since now $\Phi(x) \longrightarrow \Phi(x) - iq\theta(x)\Phi(x)$ we get

$$\begin{aligned} \partial_\mu\Phi &\rightarrow \partial_\mu(\Phi - iq\theta\Phi) = \partial_\mu\Phi - iq\partial_\mu(\theta\Phi) \\ &= \cancel{\mathcal{L}(\Phi, \partial_\mu\Phi)} + \frac{\partial\mathcal{L}}{\partial\Phi}(-iq\theta\Phi) + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\Phi)}(-iq)\partial_\mu(\theta\Phi) \\ &= (-iq\theta)\frac{\partial\mathcal{L}}{\partial\Phi}\Phi - iq\frac{\partial\mathcal{L}}{\partial(\partial_\mu\Phi)}((\partial_\mu\theta)\Phi + \theta\partial_\mu\Phi) \\ &= -iq\theta\left(\underbrace{\frac{\partial\mathcal{L}}{\partial\Phi}\Phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\Phi)}\partial_\mu\Phi}_{= 0 \text{ because } \star}\right) - iq\frac{\partial\mathcal{L}}{\partial(\partial_\mu\Phi)}(\partial_\mu\theta)\Phi \\ &= \underbrace{\frac{\partial\mathcal{L}}{\partial(\partial_\mu\Phi)}(-iq)\Phi(\partial_\mu\theta)}_{= J^\mu \text{ from } \star\star} = J^\mu(\partial_\mu\theta) = D\mathcal{L} \neq 0 \end{aligned}$$

We get

$$D\mathcal{L} = J^\mu(\partial_\mu\theta)$$

Added explanation (not in the original notes)

This is the pivot of the entire lecture. Read the boxed result slowly. It says: if you gauge a global symmetry of *any* Lagrangian — whatever its fields, whatever its interactions — the Lagrangian does not stay invariant, and the *exact* amount by which it fails is $J^\mu\partial_\mu\theta$, with J^μ the Noether current of the original global symmetry.

Three things follow immediately.

- The current appears *spontaneously*. We never had to guess it, nor to compute it in the free theory and hope it survives. It is whatever the full interacting Lagrangian's Noether current is, by construction.
- The failure is proportional to $\partial_\mu\theta$, i.e. it vanishes for constant θ — as it must, since for constant θ we are back to the exact global symmetry.
- $J^\mu\partial_\mu\theta$ has exactly the same *shape* as the term that a gauge field transforming as $A_\mu \rightarrow A_\mu + \partial_\mu\alpha$ would produce. That structural coincidence is what makes the next page work.

Historical footnote. This is the modern statement of the *gauge principle* (Weyl, 1929): a local symmetry cannot exist on its own; it *demand*s a compensating vector field. The photon is not put in by hand — it is the price of promoting θ to $\theta(x)$.

8 Combining the two transformations [p. 9*****-10*****]

4) We have the following situation:

$$D\mathcal{L} = \underbrace{J^\mu (\partial_\mu \theta)}_{\text{from } \Phi(x) \rightarrow \Phi(x) - iq\theta(x)\Phi(x)} + \underbrace{\frac{\partial \mathcal{L}}{\partial A^\mu} (\partial^\mu \alpha) + \frac{\partial \mathcal{L}}{\partial (\partial^\nu A^\mu)} (\partial^\nu \partial^\mu \alpha)}_{\text{from } A_\mu(x) \rightarrow A_\mu(x) + \partial_\mu \alpha(x)}$$

The change $\frac{\partial \mathcal{L}}{\partial A^\mu} (\partial^\mu \alpha) + \frac{\partial \mathcal{L}}{\partial (\partial^\nu A^\mu)} (\partial^\nu \partial^\mu \alpha)$ must be zero because the gauge redundancy cannot percolate into physical observables.

There is an interesting possibility. We combine the two transformations by setting $\theta(x) = \alpha(x)$,

$$\begin{cases} \Phi(x) \rightarrow \exp(-iq\alpha(x)) \Phi(x) \\ A^\mu(x) \rightarrow A^\mu(x) + \partial^\mu \alpha(x) \end{cases}$$

Combined gauge transformation

We now have

$$D\mathcal{L} = (\partial_\mu \alpha) \left(J^\mu + \frac{\partial \mathcal{L}}{\partial A_\mu} \right) + \frac{\partial \mathcal{L}}{\partial (\partial^\nu A^\mu)} (\partial^\nu \partial^\mu \alpha)$$

And the condition $D\mathcal{L} = 0$ can be obtained if we require: [p. 11*****]

$$\frac{\partial \mathcal{L}}{\partial A_\mu} = -J^\mu ; \quad \frac{\partial \mathcal{L}}{\partial (\partial^\mu A^\nu)} = -\frac{\partial \mathcal{L}}{\partial (\partial^\nu A^\mu)} \quad (\blacksquare)$$

Added explanation (not in the original notes)

How the two conditions arise. The first bracket multiplies $\partial_\mu \alpha$, so it must vanish on its own: $\partial \mathcal{L} / \partial A_\mu = -J^\mu$. The second term multiplies $\partial^\nu \partial^\mu \alpha$, which is *symmetric* in $(\mu\nu)$ because partial derivatives commute. A symmetric object contracted with an *antisymmetric* one gives zero — hence the requirement that $\partial \mathcal{L} / \partial (\partial^\mu A^\nu)$ be antisymmetric. Note that antisymmetry is enough; we do not need it to vanish.

What $\partial \mathcal{L} / \partial A_\mu = -J^\mu$ means physically. It says the photon couples to the *full* Noether current of the interacting theory — not to some current borrowed from the free theory. Written as an equation of motion (vary A_μ in $\mathcal{L} = -\frac{1}{4}F^2 + \dots$) it is precisely $\partial_\nu F^{\nu\mu} = J^\mu$: *Maxwell's equations with the correct source*. Charge conservation $\partial_\mu J^\mu = 0$ then follows from the antisymmetry of $F^{\nu\mu}$, off-shell for the matter fields.

This is Noether's second theorem. Noether's first theorem (global symmetry $\Rightarrow$ conserved current) is what gave us ★★. What is happening on this page is the second theorem: a *local* symmetry does not give a new conservation law, it gives an *identity* relating the equations of motion — here, the statement that the matter E.o.M. already imply $\partial_\mu J^\mu = 0$.

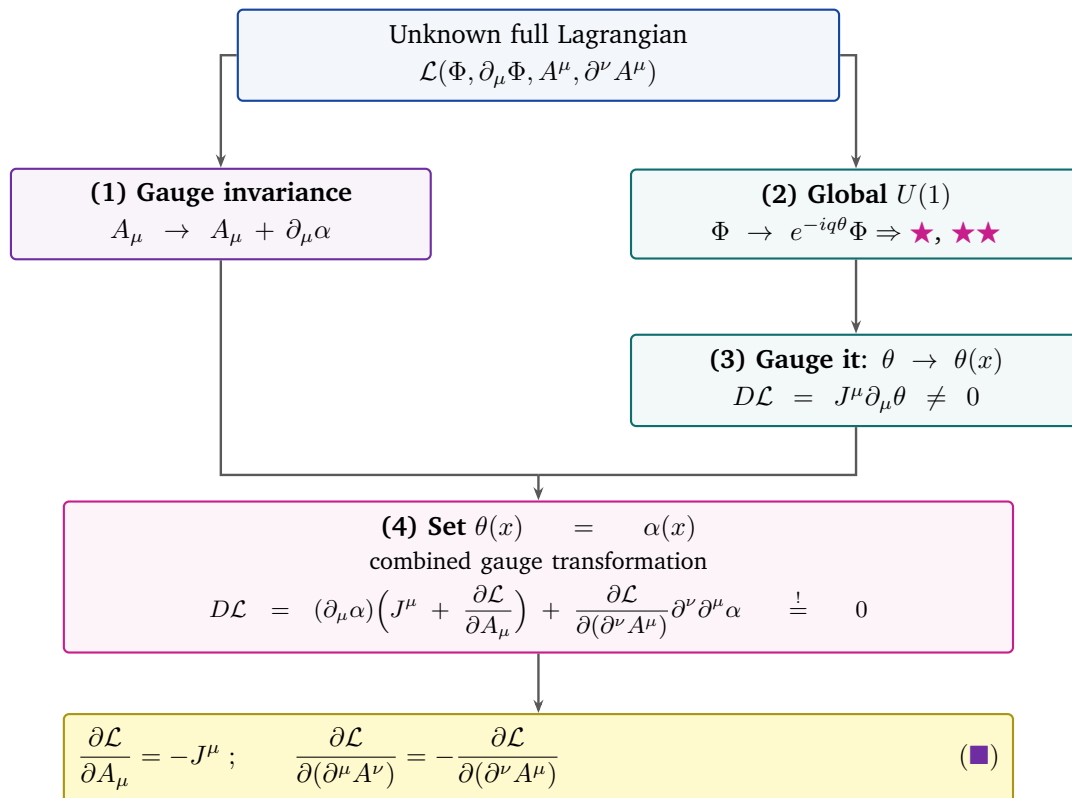


Figure 2. The logical skeleton of pp. 6*****–11*****. Two independent inputs (a gauge redundancy for A^μ , a global $U(1)$ for Φ) are welded together by the single choice $\theta = \alpha$, and the welding conditions are (■). (Added figure, not in the original notes.)

9 Comments on the conditions (■) [p. 11*****–13*****]

1) If we consider a Lagrangian such that (■) are verified, the Lagrangian will be invariant under the combined infinitesimal gauge transformation

$$\begin{cases} \Phi(x) \rightarrow \Phi(x) - iq\alpha(x)\Phi(x) \\ A_\mu(x) \rightarrow A_\mu(x) + \partial_\mu \alpha \end{cases}$$

2) Consider the second equation. If the Lagrangian depends on $\partial_\mu A_\nu$ only via $F_{\mu\nu} = -F_{\nu\mu}$, the equation is trivially satisfied

$$\mathcal{L} = \mathcal{L}(\Phi, \partial_\mu \Phi, A^\mu, F^{\mu\nu})$$

3) Suppose that we do not gauge the symmetry. Then, going back to the computation of $D\mathcal{L}$, we get [p. 12*****]

$$D\mathcal{L} = \underbrace{J^\mu (\partial_\mu \theta)} + \frac{\partial \mathcal{L}}{\partial A^\mu} (\partial^\mu \alpha) + \frac{\partial \mathcal{L}}{\partial (\partial^\nu A^\mu)} \partial^\nu \partial^\mu \alpha$$

We do not have this term anymore.

We can still have $D\mathcal{L} = 0$ if we impose

$$\frac{\partial \mathcal{L}}{\partial A^\mu} = 0 ; \quad \frac{\partial \mathcal{L}}{\partial (\partial^\mu A^\nu)} = -\frac{\partial \mathcal{L}}{\partial (\partial^\nu A^\mu)}$$

If we do not gauge the symmetry, this equation tells us that the Lagrangian cannot depend explicitly on A_μ

This is possible, but it is a limitation.

Example

Consider the Lagrangian

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\Psi}(i\gamma^\mu\partial_\mu - m)\Psi + C_e F_{\mu\nu}\bar{\Psi}\sigma^{\mu\nu}\gamma_5\Psi + C_m F_{\mu\nu}\bar{\Psi}\sigma^{\mu\nu}\Psi$$

This Lagrangian is totally acceptable, but does not feature the electromagnetic coupling $A_\mu\bar{\Psi}\gamma^\mu\Psi$ (since the Lagrangian cannot depend on A_μ), which is crucial to reproduce standard electromagnetism. [p. 13*****]

Added explanation (not in the original notes)

What those two extra operators are. With $\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu]$, the operator $F_{\mu\nu}\bar{\Psi}\sigma^{\mu\nu}\Psi$ is the **magnetic dipole moment** operator (it shifts the electron's g -factor away from 2), and $F_{\mu\nu}\bar{\Psi}\sigma^{\mu\nu}\gamma_5\Psi$ is the **electric dipole moment** operator (it violates P and T). Both are built purely from $F_{\mu\nu}$, so they satisfy the antisymmetry condition automatically and they never mention A_μ on its own — which is exactly the point being made.

Their mass dimension. $[F_{\mu\nu}] = [M]^2$ and $[\Psi] = [M]^{3/2}$, so the operators have dimension $2 + 3 = 5$. The couplings C_e, C_m therefore carry dimension $[M]^{-1}$: these are **non-renormalizable** interactions (Lecture 13's classification). They will reappear on p. 17***** as the standard example of a “non-minimal” theory.

So what exactly is lost? A theory that is only $F_{\mu\nu}$ -dependent still describes electrons that respond to electric and magnetic *fields*. What it cannot produce is the Coulomb interaction with its $1/r$ potential and the whole structure of QED: the minimal vertex $-qA_\mu\bar{\Psi}\gamma^\mu\Psi$ requires A_μ itself, which is available only if the symmetry is gauged.

Part III — The minimal coupling prescription

(pp. 13*****–18*****)

10 QED satisfies (■), and the algorithm [p. 13*****–14*****]**4) Consider the QED Lagrangian**

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\Psi}(i\gamma^\mu\partial_\mu - m)\Psi - qA_\mu\bar{\Psi}\gamma^\mu\Psi$$

This Lagrangian satisfies the criteria (■), and it is invariant under the combined finite gauge transformation.

There is a simple algorithm to make a Lagrangian consistent with (■) *and* invariant under the finite combined gauge transformation. It is called “**minimal coupling prescription**”

It works as follows.

i) We start with a free Lagrangian

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \mathcal{L}_\Phi(\Phi, \partial_\mu\Phi)$$

Free Lagrangian for Φ , endowed with a global $U(1)$ symmetry.

ii) The ordinary derivative is promoted to be a covariant derivative [p. 14*****]

$$\partial_\mu\Phi \longrightarrow D_\mu\Phi \equiv \partial_\mu\Phi + iqA_\mu\Phi$$

The resulting Lagrangian

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \mathcal{L}_\Phi(\Phi, D_\mu\Phi)$$

is a renormalizable Lagrangian that is invariant under the combined gauge transformation.

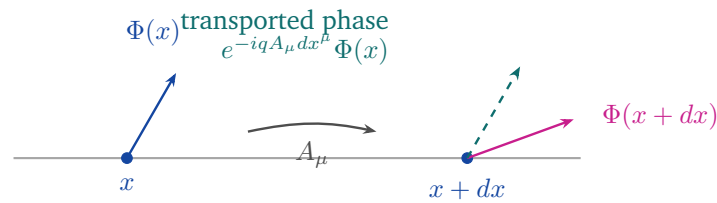
Added explanation (not in the original notes)

The algorithm in one line: write the free theory, replace every $\partial_\mu \rightarrow D_\mu = \partial_\mu + iqA_\mu$, and stop. Everything else — the interaction vertices, their relative coefficients, the conserved current, even the seagull term of scalar QED — is then fixed, with no freedom left.

Why it is guaranteed to work (the proof is the computation on p. 15*****). The only obstacle to gauge invariance was that $\partial_\mu\Phi$ transforms *badly*: differentiating $e^{-iq\alpha(x)}$ produces an unwanted $\partial_\mu\alpha$ term. D_μ is engineered so that the $+iqA_\mu\Phi$ piece produces *exactly* the opposite term when $A_\mu \rightarrow A_\mu + \partial_\mu\alpha$. The two errors cancel, and $D_\mu\Phi$ transforms as nicely as Φ itself.

Sign bookkeeping. The relative sign inside D_μ is tied to the sign convention in $\Phi \rightarrow e^{-iq\alpha}\Phi$ and $A_\mu \rightarrow A_\mu + \partial_\mu\alpha$. Change any one of the three and the other two must follow. With the conventions of these notes: $\Phi \rightarrow e^{-iq\alpha}\Phi$, $A_\mu \rightarrow A_\mu + \partial_\mu\alpha$, $D_\mu = \partial_\mu + iqA_\mu$.

Where the covariant derivative comes from geometrically (not in the notes). A gauge transformation rotates the phase of Φ *differently at each point*. Comparing $\Phi(x)$ with $\Phi(x + dx)$ therefore compares two quantities in two different “phase frames” — the difference is not meaningful. A_μ is the *connection* that tells you how to transport the phase from one point to the next, and D_μ is the derivative that first transports, then subtracts. This is the same idea as the covariant derivative of general relativity, with the Christoffel symbols replaced by iqA_μ .



$$D_\mu \Phi = (\text{value here}) - (\text{value there, transported here}) = \partial_\mu \Phi + iqA_\mu \Phi$$

Figure 3. The covariant derivative as a *comparator*. An ordinary derivative subtracts phases defined in different frames, which is why it transforms badly; A_μ supplies the transport that makes the subtraction meaningful. (Added figure, not in the original notes.)

11 Example: scalar QED [p. 14*****-15*****]

We start from

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \underbrace{(\partial_\mu \Phi^\dagger)(\partial^\mu \Phi) - m^2 \Phi^\dagger \Phi}$$

Invariant under a global $U(1)$ transformation $\Phi \rightarrow \exp(-iq\alpha)\Phi$.

In particular, notice that

$$\partial_\mu \Phi \longrightarrow \partial_\mu (e^{-iq\alpha}\Phi) = e^{-iq\alpha}(\partial_\mu \Phi)$$

and, consequently, we have

$$(\partial_\mu \Phi^\dagger)(\partial^\mu \Phi) \longrightarrow e^{+iq\alpha}(\partial_\mu \Phi^\dagger)e^{-iq\alpha}(\partial^\mu \Phi) = (\partial_\mu \Phi^\dagger)(\partial^\mu \Phi)$$

If we gauge the symmetry, the term $\Phi^\dagger \Phi$ is still invariant. [p. 15*****]

However, the derivative gives problems since

$$\partial_\mu \Phi \longrightarrow \partial_\mu (e^{-iq\alpha(x)}\Phi) = -iq(\partial_\mu \alpha)e^{-iq\alpha}\Phi + e^{-iq\alpha}(\partial_\mu \Phi)$$

and $(\partial_\mu \Phi^\dagger)(\partial^\mu \Phi)$ is no longer invariant.

↪ **The covariant derivative fixes the problem. We have:**

$$\begin{aligned} D_\mu \Phi &\equiv \partial_\mu \Phi + iqA_\mu \Phi \longrightarrow \partial_\mu (e^{-iq\alpha}\Phi) + iq(A_\mu + \partial_\mu \alpha)e^{-iq\alpha}\Phi \\ &= \cancel{(-iq)(\partial_\mu \alpha)e^{-iq\alpha}\Phi} + iqA_\mu e^{-iq\alpha}\Phi + \cancel{iq(\partial_\mu \alpha)e^{-iq\alpha}\Phi} \\ &\quad + e^{-iq\alpha}(\partial_\mu \Phi) \\ &= e^{-iq\alpha}(\partial_\mu + iqA_\mu)\Phi = e^{-iq\alpha} D_\mu \Phi \end{aligned}$$

$$D_\mu \Phi \longrightarrow e^{-iq\alpha(x)} D_\mu \Phi$$

When the global symmetry is gauged, the covariant derivative has the same transformation property as the ordinary derivative does when the symmetry is global.

Consequently, $(D_\mu \Phi)^\dagger (D^\mu \Phi)$ is trivially invariant.

Added explanation (not in the original notes)

The two cancelling terms, spelled out. In the middle line above, the first term $(-iq)(\partial_\mu\alpha)e^{-iq\alpha}\Phi$ comes from differentiating the exponential; the third term $+iq(\partial_\mu\alpha)e^{-iq\alpha}\Phi$ comes from the shift $\partial_\mu\alpha$ in A_μ . They are equal and opposite. *This single cancellation is the whole content of the gauge principle:* the inhomogeneous transformation of the gauge field was designed — back on p. 6***** — to be exactly what is needed to absorb the derivative of the phase.

Note the structure of the result. $D_\mu\Phi$ transforms *homogeneously*, i.e. it picks up the same phase factor as Φ itself, with no extra term. Any expression built from Φ and $D_\mu\Phi$ in which every Φ is paired with a $\Phi^\dagger$ is therefore automatically invariant — and this is true for *finite* α , not just infinitesimal ones. That is why p. 13***** could promise invariance under the *finite* combined gauge transformation, which conditions (■) alone (being infinitesimal statements) do not guarantee.

12 The scalar QED Lagrangian and its current [p. 16*****-17*****]

The final Lagrangian is

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + (D_\mu\Phi)^\dagger(D^\mu\Phi) - m^2\Phi^\dagger\Phi$$

Scalar QED

Let us now check the conditions (■). First, we write the Lagrangian as follows

We write explicitly the covariant derivatives.

$$\begin{aligned}\mathcal{L} &= -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + (\partial_\mu\Phi^\dagger)(\partial^\mu\Phi) - m^2\Phi^\dagger\Phi && \longrightarrow \text{free theory} \\ &\quad -iqA_\mu\left[\Phi^\dagger(\partial^\mu\Phi) - (\partial^\mu\Phi^\dagger)\Phi\right] + q^2A_\mu A^\mu\Phi^\dagger\Phi && \longrightarrow \text{interactions}\end{aligned}$$

The conserved current J^μ associated to the $U(1)$ global symmetry is given by:

$$\begin{aligned}J^\mu &= \frac{\partial\mathcal{L}}{\partial(\partial_\mu\Phi)}D\Phi + D\Phi^\dagger\frac{\partial\mathcal{L}}{\partial(\partial_\mu\Phi^\dagger)} && \begin{cases} D\Phi = -iq\Phi \\ D\Phi^\dagger = +iq\Phi^\dagger \end{cases} \\ &= \left[(\partial^\mu\Phi^\dagger) - iqA^\mu\Phi^\dagger\right](-iq)\Phi + iq\Phi^\dagger\left[\partial^\mu\Phi + iqA^\mu\Phi\right] \\ &= iq\left[\Phi^\dagger(\partial^\mu\Phi) - (\partial^\mu\Phi^\dagger)\Phi\right] - 2q^2A^\mu\Phi^\dagger\Phi\end{aligned}$$

[p. 17*****]

$$J^\mu = iq\left[\Phi^\dagger(\partial^\mu\Phi) - (\partial^\mu\Phi^\dagger)\Phi\right] - 2q^2A^\mu\Phi^\dagger\Phi$$

↔ Note that this is a case in which the conserved current **depends on** A^μ .

If we compute $\partial\mathcal{L}/\partial A_\mu$, we indeed check that

$$\frac{\partial\mathcal{L}}{\partial A_\mu} = -iq\left[\Phi^\dagger(\partial^\mu\Phi) - (\partial^\mu\Phi^\dagger)\Phi\right] + 2q^2 A^\mu\Phi^\dagger\Phi = -J^\mu, \quad \text{as expected.}$$

Added explanation (not in the original notes)

Everything promised on p. 5***** has just been delivered. Compare:

naive coupling, p. 5*****

$$-iqA_\mu[\Phi^\dagger\partial^\mu\Phi - (\partial^\mu\Phi^\dagger)\Phi] \text{ only}$$

minimal coupling, p. 16*****

$$\text{the same term plus } q^2 A_\mu A^\mu \Phi^\dagger\Phi$$

the failure, p. 5*****

$$\partial_\mu J^\mu = 2q^2 \partial_\mu (A^\mu \Phi^\dagger\Phi)$$

the repaired current

$$J^\mu \rightarrow J^\mu - 2q^2 A^\mu \Phi^\dagger\Phi$$

The extra term the prescription forces on the Lagrangian, and the extra term it forces into the current, are the *same* $q^2 A^\mu \Phi^\dagger\Phi$ structure that was missing. Nothing was guessed.

The compact form. The boxed current is nothing but

$$J^\mu = iq\left[\Phi^\dagger(D^\mu\Phi) - (D^\mu\Phi)^\dagger\Phi\right],$$

i.e. the free-theory current with $\partial^\mu \rightarrow D^\mu$. (Expand it: the iqA^μ pieces add rather than cancel and give $-2q^2 A^\mu \Phi^\dagger\Phi$.) In this form conservation is a two-line check: using $\partial_\mu(\Phi^\dagger D^\mu\Phi) = (D_\mu\Phi)^\dagger D^\mu\Phi + \Phi^\dagger D_\mu D^\mu\Phi$ and the E.o.M. $D_\mu D^\mu\Phi = -m^2\Phi$, everything cancels and $\partial_\mu J^\mu = 0$ — on the E.o.M. of the fully interacting theory, which is what example (ii) failed to achieve.

The $q^2 A^2 \Phi^\dagger\Phi$ term is physical. It is the “seagull” vertex: two photons attaching to the scalar line at the same point. It has no analogue in spinor QED (the Dirac Lagrangian is linear in derivatives, so D_μ appears only once), which is one more way of saying that spinor QED was the exceptional case all along.

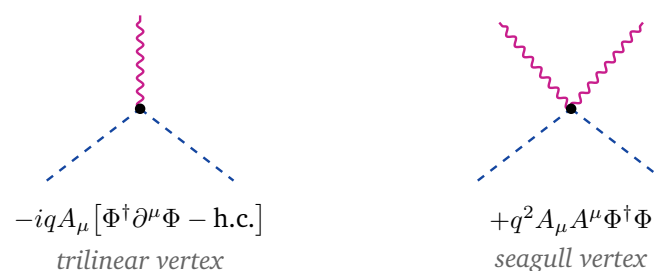


Figure 4. The two interaction vertices produced by $\partial_\mu \rightarrow D_\mu$ in scalar QED (dashed = scalar, wavy = photon). The naive “couple to the free current” recipe produces only the first; gauge invariance requires both, with a fixed relative coefficient. (Added figure, not in the original notes.)

13 Comment: why “minimal”? [p. 17*****–18*****]

We call it “minimal coupling” because it introduces the **smallest possible modification to the Lagrangian** necessary to ensure that the theory is consistent with local gauge invariance at the renormalizable level.

Consider, for instance, the theory

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\Psi}(i\gamma^\mu\partial_\mu - m)\Psi \underbrace{- qA_\mu\bar{\Psi}\gamma^\mu\Psi}_{\text{renormalizable interaction}} \\ + \underbrace{C_m F_{\mu\nu}\bar{\Psi}\sigma^{\mu\nu}\Psi + C_e F_{\mu\nu}\bar{\Psi}\sigma^{\mu\nu}\gamma_5\Psi}_{\text{non-renormalizable interactions}}$$

This theory is perfectly consistent, but “non-minimal” because of the presence of non-renormalizable interactions. [p. 18*****]

Added explanation (not in the original notes)

“Minimal” is a choice, not a theorem. The Lagrangian above satisfies (■) just as well as QED does — the dipole operators are built from $F_{\mu\nu}$ and are gauge invariant on their own. Gauge invariance therefore does *not* single out QED; it single out a *family*. What picks the minimal member is the extra demand of **renormalizability**: keep only operators of mass dimension ≤ 4 .

Modern reading. Today we would say the non-minimal terms are perfectly legitimate *effective* interactions, suppressed by powers of a high scale Λ : $C_m \sim 1/\Lambda$. Measuring C_m is measuring the electron’s anomalous magnetic moment; the striking fact is that in QED it is not a free parameter at all but is *predicted* by loops, $C_m \propto \alpha/(4\pi m)$, in agreement with experiment to twelve digits. Minimal coupling is thus not merely economical — it is experimentally right.

14 Extra comment: on-shell vs off-shell gauge invariance [p. 18*****]

Going back to the initial discussion, a **third limitation** of

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - A_\mu J^\mu$$

is that gauge invariance is only valid on-shell, meaning that

$$D\mathcal{L} = -(\partial_\mu\alpha)J^\mu = -\partial_\mu(\alpha J^\mu) + \alpha(\partial_\mu J^\mu)$$

reduces to a total derivative only if $\partial_\mu J^\mu = 0$, which is guaranteed on the E.o.M.

↪ From a broader perspective, we want gauge invariance to be valid also **off-shell**, in order to be considered a **fundamental redundancy in the formulation of a theory, not just a property of its solutions**.

For this reason, we assume $D\mathcal{L} = 0$ as a consequence of a gauge transformation.

Added explanation (not in the original notes)

Why this matters more than it looks. “On-shell” means: only for field configurations that solve the equations of motion. But in the quantum theory — in the path integral, or in the canonical formalism with off-shell intermediate states — one integrates over *all* field configurations, not just solutions. A symmetry that only holds on solutions is therefore not available where it is needed most: it cannot be used to remove the unphysical polarizations of A_μ consistently, and the Ward identities that guarantee unitarity would fail off-shell.

Three limitations, one cure. The lecture opened with two limitations and closes with a third; all three are cured by the same construction:

Limitation	Cured by
J^μ assumed independent of A_μ	the gauged current <i>may</i> depend on A_μ (scalar QED, p. 17*****)
no recipe for constructing J^μ	the Noether current of the gauged symmetry, produced automatically (p. 9*****)
gauge invariance only on-shell	$D\mathcal{L} = 0$ identically, by construction (p. 11*****)

Summary of Lecture 15 — the logic in six lines

1. “Couple the photon to a conserved current” is not a construction: the current of the free theory need not remain conserved once the coupling changes the E.o.M. (spinor QED gets away with it; scalar QED does not).
2. Start instead from a free theory with a *global* $U(1)$ symmetry, and promote the parameter to $\theta(x)$.
3. The Lagrangian then fails to be invariant by exactly $D\mathcal{L} = J^\mu \partial_\mu \theta$, with J^μ the Noether current of the full theory.
4. Identify $\theta(x)$ with the gauge parameter $\alpha(x)$ of $A_\mu \rightarrow A_\mu + \partial_\mu \alpha$. Invariance then requires $\frac{\partial \mathcal{L}}{\partial A_\mu} = -J^\mu$ and antisymmetry of $\frac{\partial \mathcal{L}}{\partial(\partial^\mu A^\nu)}$ — conditions (■).
5. Both are automatically satisfied if $\mathcal{L}$ depends on ∂A only through $F_{\mu\nu}$ and on $\partial\Phi$ only through $D_\mu\Phi = \partial_\mu\Phi + iqA_\mu\Phi$. That substitution *is* the minimal coupling prescription.
6. Result: interactions, the conserved current, and the seagull term all come out fixed, and gauge invariance holds off-shell and for finite α .

Looking ahead. Everything on these eighteen pages used the abelian group $U(1)$, where phases commute. Replacing $e^{-iq\alpha}$ by a non-commuting matrix $U(x)$ turns $D_\mu = \partial_\mu + iqA_\mu$ into $D_\mu = \partial_\mu + igA_\mu^a T^a$ and forces $F_{\mu\nu}$ to acquire a term $\propto [A_\mu, A_\nu]$ — the gauge field then carries charge and interacts with itself. That is Yang–Mills theory, and the logic of Part II carries over verbatim.